

Neel Agarwal

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SKILLS

Programming & Scripting: Python, SQL, PySpark, R, Java, C/C++, JavaScript, Bash/Shell, Boto3

Data Engineering & Cloud Platforms: ETL/ELT Pipelines, Apache Spark, Apache Airflow, Apache Kafka, dbt, Databricks, AWS (S3, EC2, Lambda, Glue, Athena, Redshift, RDS, EMR, IAM, EventBridge, API Gateway, SQS), Data Modeling (Star Schema), Data Warehousing, Distributed Computing, System Design, CI/CD, Data Governance

ML & Data Science: Pandas, NumPy, PyTorch, TensorFlow, Keras, MLflow, Prophet, SHAP, NLTK, NetworkX, CNN, RNN

Data Visualization & Dev Tools: Tableau, QuickSight, Jupyter, Git, Docker, Plotly, Matplotlib, Seaborn, CRM, A/B Testing

EXPERIENCE

Data Engineer, Wearable Technology (Healthcare)

Jun 2024 - Present

University of Rochester Medical Center

Rochester, NY

- Engineered scalable data ingestion pipelines using **Python**, **SQL**, and **AWS S3** to process 40 hours of raw ECG data per subject from Frontier X2 devices, reducing data preparation time by 25% and ensuring high-throughput ingestion across data sources.
- Designed and orchestrated automated transformation workflows with **AWS Glue**, **dbt**, and **Apache Airflow**, enabling consistent preprocessing and harmonization of physiological signals and improving data readiness for machine learning by 40%.
- Built **data integrity** and **validation** frameworks using Signal Quality Indices (SQIs), **PySpark**, and statistical thresholds, reducing noise and missing value artifacts, and boosting downstream model reliability by 35%.
- Implemented secure and governed data storage using Snowflake and RBAC, with **CI/CD** pipelines containerized using **Docker** and version-controlled in **Git**, ensuring reproducible ETL deployments and **HIPAA**-compliant data workflows.
- Integrated real-time anomaly detection pipelines with **Spark Streaming** and AWS Lambda, deploying on **EMR** to detect cardiac events from ECG data with 90% model accuracy, reducing latency and enhancing response capability.
- Delivered stakeholder-facing analytics by integrating **Tableau dashboards** with curated Snowflake datasets, enabling clinicians and researchers to track patient metrics, signal health, and model performance in real time.

Machine Learning Engineer, Customer Retention (Retail)

Sep 2024 - Dec 2024

Wegmans Food Markets

Rochester, NY

- Built and automated scalable ML pipelines using **Python** (pandas, NumPy), **SQL**, and **MLflow** to process and transform transaction logs from ~300K Meals2GO users, reducing lag in feature delivery for ML models and dashboard reporting.
- Developed time-series forecasting models using **Prophet** and **XGBoost** to segment and predict customer engagement trends, enabling timely and personalized outreach through CRM analytics.
- Achieved a **20% boost in customer retention** (~360K users) by translating ML predictions into actionable promotions, translating ML insights into repeat purchases and measurable sales growth.

Analytics Engineer, Retail Sales (Oil & Gas)

Jan 2021 - Mar 2023

Indian Oil Corporation Ltd.

Gangtok, India

- Automated retail sales data transformation using **MySQL**, **Python**, and **dbt** on **Databricks**, reducing ETL processing time by 25% and ensuring consistent, governed pipelines.
- Built **predictive inventory models** and implemented pricing strategies, reducing operational losses by 20% (~\$6.6K savings) through optimized stock and purchase timing.
- Developed **Tableau** dashboards on Snowflake datasets using **statistical analysis** and **A/B testing** to visualize fuel sales, inventory, and pricing trends, enabling regional managers to make faster, data-informed decisions and optimize promotional strategies.
- Presented analytical insights** to both technical and non-technical stakeholders, translating statistical findings into actionable business strategies and improving collaboration across sales, finance, and logistics teams.

PROJECTS

Marketing Analytics Pipeline on AWS for Calendly KPI Reporting

- Engineered a serverless data pipeline on AWS using Lambda, API Gateway, S3, and AWS Glue (Crawlers, Data Catalog, ETL Jobs), with Athena and QuickSight, to automate KPI reporting from Calendly and marketing data, improving spend allocation and marketing ROI visibility through SQL-accessible datasets and dynamic dashboards.

Real-Time CRM Lead Processing & Notification System

- Built a real-time serverless end-to-end pipeline on AWS (Lambda, S3, SQS, API Gateway) to instantly capture and process CRM webhooks for new sales leads, enrich them with ownership details, and alert the marketing/sales teams using Slack, cutting manual work and speeding up lead response times by 95%.

Databricks Pipeline for Tweet Classification via Spark and MLflow

- Designed a real-time sentiment analysis pipeline on Databricks using Apache Spark Streaming and the Medallion architecture, processing over 200K tweets from Amazon S3, and integrated a pre-trained HuggingFace Transformer model deployed via MLflow for scalable, low-latency classification, enabling faster access to customer sentiment for timely marketing decisions.

EDUCATION

- M.S. Data Science, University of Rochester, NY, United States (GPA: 3.92) Dec 2024
- B. Tech. (Computer Science & Engineering), LP University, Punjab, India (GPA: 3.68) Jun 2021